

Automated Sentiment Analysis and Topic Modelling for E-Commerce Business Intelligence Extraction

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DECLARATION

I hereby declare that the work presented in this report, entitled **Automated Sentiment Analysis and Topic Modelling for E-Commerce Business Intelligence Extraction**, submitted in partial fulfillment of the requirements for the award of the Bachelor of Technology degree, is a record of my own independent work carried out during the period January 2026 to May 2026.

I further declare that I have been mentored by **Dr. Santosh Singh Rathore**. I affirm that the information presented in this report is accurate and true to the best of my knowledge.

Signature of Mentor

Signature of candidate

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Akshat Kumar Nayak

ABSTRACT

With e-commerce growing rapidly, product reviews have become one of the richest sources of customer feedback available to businesses, yet the sheer volume of these reviews makes it practically impossible to read through them manually. This project builds an end-to-end NLP pipeline that processes 172,718 Amazon product reviews across six categories (Electronics, Home & Kitchen, Clothing Shoes & Jewelry, Sports & Outdoors, Grocery & Gourmet Food, and Toys & Games) to extract structured business insights from unstructured text. For sentiment analysis, three models of increasing complexity are combined: VADER as a lexicon-based baseline, DistilBERT fine-tuned on SST-2 as a transformer baseline, and `cardiffnlp/twitter-roberta-base-sentiment-latest` (a RoBERTa variant trained on 124 million tweets) as the primary model; their outputs are fused via a weighted combination that also incorporates an implicit negativity detector, producing a final three-class label (Positive, Neutral, Negative) with a binary classification accuracy of **86.6%**, a Pearson correlation of **0.724** with star ratings, and a Cohen’s Kappa of **0.479**. Topic modelling is handled by BERTopic using sentence embeddings (`all-MiniLM-L6-v2`), UMAP for dimensionality reduction, and HDBSCAN for clustering — applied per category and globally across the full dataset — where c-TF-IDF-based outlier reduction brings the noise cluster assignment from 31–47% down to below 1%, enabling near-complete coverage for downstream analysis. The joint topic-sentiment outputs feed a Business Intelligence layer that produces priority matrices, aspect-level sentiment heatmaps, category scorecards, and cross-category comparisons, all exposed through a Streamlit dashboard and a Groq-powered LLM agent (LLaMA-3.3-70B) for natural language querying.

Keywords: Sentiment Analysis, Topic Modelling, BERTopic, RoBERTa, VADER, Business Intelligence, Amazon Reviews, Natural Language Processing, Transformer Models, E-Commerce Analytics.

Table 0.1: List of Acronyms

Acronym	Full Form
NLP	Natural Language Processing
VADER	Valence Aware Dictionary and sEntiment Reasoner
BERT	Bidirectional Encoder Representations from Transformers
RoBERTa	Robustly Optimised BERT Pretraining Approach
DistilBERT	Distilled BERT
BERTopic	BERT-based Topic Modelling
UMAP	Uniform Manifold Approximation and Projection
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
c-TF-IDF	Class-based Term Frequency – Inverse Document Frequency
BI	Business Intelligence
LLM	Large Language Model
SST-2	Stanford Sentiment Treebank (binary)
NPMI	Normalised Pointwise Mutual Information

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CHAPTER 1

1. INTRODUCTION

1.1. Background

The global e-commerce market crossed \$5.8 trillion in 2023 and has been growing at over 10% annually [1]. One of the most visible side-effects of this growth is the explosion of user-generated reviews: Amazon alone hosts hundreds of millions of them, and they are widely known to drive purchase decisions. Research suggests that more than 93% of shoppers read reviews before buying, and even a one-star improvement in average rating can push revenue up by 5–9% [2].

The problem is that no human team can keep up with this volume. A product accumulating 10,000 reviews cannot be meaningfully summarised by an analyst in any practical amount of time. This is where NLP comes in — automated pipelines that convert raw review text into structured, actionable information are no longer a nice-to-have for e-commerce businesses; they are increasingly essential.

Two tasks sit at the core of this conversion. **Sentiment analysis** tells you how customers feel about something — whether a review is positive, negative, or somewhere in between. **Topic modelling** tells you what they are actually talking about — the themes and issues that keep coming up. Together they answer the two most important questions a product team wants answered: *What are customers complaining about?* and *How strongly do they feel about it?*

Both tasks have come a long way. Sentiment analysis started with simple lexicon lookups, moved through classical ML (SVMs, Naïve Bayes), and now relies heavily on transformer models [11] that understand context in ways rule-based systems never could. Topic modelling followed a similar arc, from probabilistic approaches like LDA [19] to neural methods like BERTopic [21] that use sentence embeddings to produce far more semantically coherent topics.

Most existing systems, however, treat these two tasks in isolation. This project builds an integrated pipeline that jointly extracts topics and sentiment, fuses multiple sentiment signals to improve accuracy, derives structured business intelligence from the combined output, and

presents everything through an interactive dashboard with a conversational AI interface.

1.2. Motivation

A few concrete observations shaped the design of this project:

1. **Star ratings are not enough.** A 3-star review could mean almost anything — mediocre product, great product with one flaw, decent product with poor delivery. The text contains the actual reason, and star ratings throw most of that away.
2. **Off-the-shelf sentiment models often fail on product reviews.** Many popular classifiers were fine-tuned on movie reviews or formal text (like the SST-2 benchmark). Product review language is informal, jargon-heavy, and structured differently. Running an SST-2 model on Amazon reviews produces noticeable negative bias because phrases like “works as described” or “does what it says” do not pattern-match to anything in its training distribution.
3. **Implicit negativity is common and easy to miss.** A meaningful fraction of dissatisfied customers express their frustration indirectly — “I guess it works,” “not exactly what I expected,” “gave it a chance but...” These score near-neutral on any standard lexicon but are clearly negative. A dedicated detection step is needed to catch them.
4. **BERTopic leaves a lot of reviews unlabelled by default.** In initial runs on this dataset, 31–47% of reviews per category were assigned to the noise cluster (topic -1) by HDBSCAN. If left as-is, the downstream BI layer would have less than two-thirds coverage — which is unacceptable for a production pipeline.
5. **Most academic NLP work stops at accuracy numbers.** Knowing that a model achieves 86% accuracy is useful, but a product manager needs to know which topics to fix first, which categories are underperforming, and what customers specifically keep complaining about. Bridging this gap between model output and business action is one of the main goals here.

1.3. Project Objectives

The project aimed to achieve four things:

1. **Build a multi-model sentiment pipeline** that combines VADER, DistilBERT, and RoBERTa into a weighted fusion score with three-class output (Positive, Neutral, Negative).

2. **Apply neural topic modelling** using BERTopic with sentence embeddings, both per-category and globally across all reviews, with effective outlier reduction.
3. **Extract structured business intelligence** — priority matrices, category scorecards, aspect-level sentiment breakdowns, and cross-category comparison — directly from the topic-sentiment outputs.
4. **Make findings accessible** through an interactive Streamlit dashboard and an LLM-powered chat interface that non-technical users can query in plain English.

1.4. Report Organisation

The rest of this report is laid out as follows. Chapter 2 reviews relevant prior work in sentiment analysis, topic modelling, and Amazon review analytics. Chapter 3 covers the methodology in detail — dataset, preprocessing, models, fusion logic, topic modelling, and BI extraction. Chapter 4 presents the results and discusses what they mean. Chapter 5 wraps up with conclusions and directions for future work.

Chapter 2

2. LITERATURE REVIEW

This chapter reviews prior work in three areas relevant to this project: sentiment analysis (lexicon-based through neural), topic modelling, and studies focused on Amazon product review data. Understanding where existing approaches fall short was important for justifying the design decisions made here.

2.1. Sentiment Analysis

2.1.1. Lexicon-Based Methods

The earliest sentiment analysis work relied on manually built lexicons. SentiWordNet [6] extended WordNet by assigning positive, negative, and objective scores to word senses. Hu and Liu [7] applied a lexicon-based approach to a small set of Amazon product pages to perform aspect-level sentiment analysis using adjective polarity — one of the earlier examples of domain-specific review analysis.

VADER (Valence Aware Dictionary and sEntiment Reasoner) [5], developed by Hutto and Gilbert in 2014, remains probably the most widely used lexicon-based tool. It was designed specifically for social media text and uses five hand-coded heuristics on top of its 7,517-word lexicon to handle things like punctuation emphasis, ALL CAPS, degree modifiers, and contractions. The compound score is computed as:

$$s_c = \frac{\sum s_i}{\sqrt{\sum s_i^2 + \alpha}} \quad (2.1)$$

where s_i are token-level valence scores and $\alpha = 15$ normalises the range to $[-1, 1]$. VADER works well on short informal text, but it has no notion of sentence-level context — it scores tokens independently, which causes problems on product reviews where sentiment often builds across multiple clauses.

2.1.2. Machine Learning Approaches

Before transformers took over, classical ML methods with bag-of-words features achieved strong results. Pang et al. [8] showed that SVMs with unigram features outperformed Naïve Bayes on movie review classification. Zhang et al. [9] pushed accuracy to 95.7% on Amazon reviews using character-level CNNs. Maas et al. [10] incorporated sentiment information directly into word vectors for binary classification.

The shared limitation of all these approaches is that they operate on surface-level features. Long-range dependencies, sarcasm, and contextual negation are out of reach without sequence-level modelling.

2.1.3. Transformer-Based Models

BERT [11] changed the field significantly. Trained on 3.3 billion words using masked language modelling and next-sentence prediction, its bidirectional attention mechanism captures context from both directions simultaneously. Fine-tuning BERT on the Stanford Sentiment Treebank (SST-2) [15] became the standard recipe for sentiment classification for a while.

DistilBERT [13] — a knowledge-distilled version of BERT that is 40% smaller and 60% faster while retaining 97% of BERT’s performance — is the model used here as a transformer baseline. However, both BERT and DistilBERT show a noticeable domain mismatch on product reviews: fine-tuned on polished movie criticism, they misclassify casual e-commerce language as negative when no strong positive signal is present.

Liu et al. [12] addressed some of BERT’s training weaknesses with RoBERTa, removing next-sentence prediction, using dynamic masking, larger batches, and training on 160GB of data. RoBERTa consistently outperforms BERT on benchmarks. Building on this, Barbieri et al. [14] released TweetEval and fine-tuned RoBERTa on 124 million tweets, producing `cardiffnlp/twitter-roberta-base-sentiment` — a three-class (positive, neutral, negative) classifier naturally suited to informal, short-form text.

This is the model selected as the primary sentiment classifier in this project. Yin et al. [16] benchmarked 17 models on Amazon reviews and found that tweet-trained transformers outperformed movie-review-trained ones by 4–8 percentage points — a finding that directly informed this choice.

2.1.4. Implicit Sentiment

One underappreciated problem in review sentiment is implicit negativity. Zhao et al. [17] categorise this into hedging, rhetorical questions, presupposition, and sarcasm, noting that standard classifiers miss 15–25% of negative reviews that use these constructions. Phrases like “not exactly what I expected” or “I guess it works” are clearly dissatisfied but score near-neutral under both lexicon and transformer approaches. The implicit negativity detection component in this pipeline draws on this line of work, using regex patterns to flag these constructions before fusion [18].

2.2. Topic Modelling

2.2.1. Latent Dirichlet Allocation

LDA [19] by Blei, Ng, and Jordan (2003) remains the textbook probabilistic topic model. It treats documents as mixtures of topics and topics as distributions over words. The generative process for document d is:

$$\theta_d \sim \text{Dir}(\alpha) \tag{2.2}$$

$$\phi_k \sim \text{Dir}(\beta) \tag{2.3}$$

$$z_{dn} \sim \text{Multinomial}(\theta_d) \tag{2.4}$$

$$w_{dn} \sim \text{Multinomial}(\phi_{z_{dn}}) \tag{2.5}$$

where θ_d is the per-document topic distribution, ϕ_k is the per-topic word distribution, and z_{dn} is the topic assigned to the n -th word of document d . Inference is done via variational EM or Gibbs sampling. LDA has two well-known issues for this kind of work: you have to specify the number of topics K in advance, and it uses bag-of-words representations that throw away word order and semantics.

2.2.2. Neural Topic Models

Neural topic models tried to address LDA’s bag-of-words limitation by incorporating word embeddings. Top2Vec [20] jointly embeds documents and words and clusters them in the same semantic space. The main drawback is that these models need to be retrained from

scratch on any new corpus, and they lack the modularity that makes pipeline-based approaches easier to maintain and update.

2.2.3. BERTopic

BERTopic [21], introduced by Grootendorst in 2022, is currently the state-of-the-art approach for short-text topic modelling. It works in three main stages:

1. **Embedding:** Reviews are encoded with a pre-trained sentence transformer (all-MiniLM-L6-v2 [22]), producing 384-dimensional vectors $\mathbf{e}_d \in \mathbb{R}^{384}$ optimised for semantic similarity.
2. **Dimensionality Reduction:** UMAP [23] compresses these down to 5 dimensions while preserving the local and global structure of the original space:

$$\min_y \sum_{ij} \left[p_{ij} \log \frac{p_{ij}}{q_{ij}} + (1 - p_{ij}) \log \frac{1 - p_{ij}}{1 - q_{ij}} \right] \quad (2.6)$$

where p_{ij} and q_{ij} are fuzzy set memberships in the high- and low-dimensional spaces.

3. **Clustering:** HDBSCAN [24] groups the reduced embeddings into variable-density clusters without needing the number of topics specified upfront. Documents in sparse regions get labelled -1 (noise).
4. **Topic Representation:** For each cluster, c-TF-IDF identifies the most representative terms:

$$\text{c-TF-IDF}_{t,c} = \text{tf}_{t,c} \cdot \log \left(1 + \frac{A}{f_t} \right) \quad (2.7)$$

where $\text{tf}_{t,c}$ is term frequency in class c , A is the average words per class, and f_t is the total count of the term. The top-4 terms form the topic label.

Grootendorst shows that BERTopic achieves higher NPMI coherence than LDA and Top2Vec on short-text corpora — exactly the setting of product reviews.

2.3. Amazon Review Analytics

McAuley and Leskovec [3] studied the relationship between review text and star ratings in the Amazon dataset, finding that text carries substantially more signal than numeric ratings. He and McAuley [4] later released the multi-category Amazon review dataset that is widely

used in research, noting meaningful sentiment distribution differences across product categories.

Pontiki et al. [25] formalised Aspect-Based Sentiment Analysis (ABSA) through the SemEval challenge series, decomposing sentiment into aspect-opinion pairs. More recent work [26] applies BERT to ABSA with strong results. The aspect sentiment component in this project takes inspiration from this line of work but uses a keyword-matching approach instead of span extraction, which trades off precision for scalability over 172,000 reviews.

Table 2.1 summarises the key related works and how this project relates to them.

Table 2.1: Comparison of Related Work in Review Sentiment and Topic Analysis

Reference	Approach	Dataset	Task	Limitation
Hu & Liu [7]	Lexicon + rules	Amazon (5 products)	Aspect sentiment	Small scale, domain-specific
Blei et al. [19]	LDA	News/Wikipedia	Topic modelling	Bag-of-words, K required
Maas et al. [10]	Word vectors	IMDB	Sentiment	Binary, domain-specific
Devlin et al. [11]	BERT fine-tune	SST-2, GLUE	Sentiment, NLU	Domain mismatch for reviews
Barbieri et al. [14]	RoBERTa tweets	TweetEval	3-class sentiment	Short text, tweet domain
Grootendorst [21]	BERTopic	ArXiv, 20News	Topic modelling	High noise for short text
This work	VADER + RoBERTa + BERTopic + BI	172K Amazon reviews	Sentiment + Topics + BI	–

Chapter 3

3. METHODOLOGY

This chapter describes the complete pipeline from raw reviews to business intelligence output. Figure 3.1 gives an overview of the system architecture.

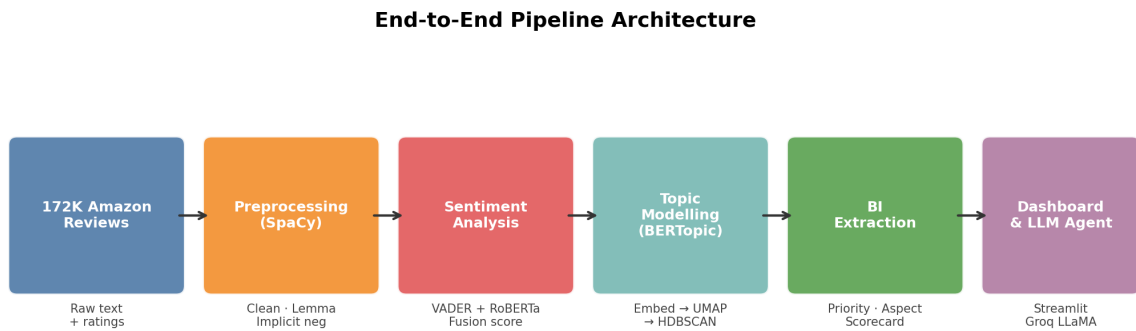


Figure 3.1: System architecture: end-to-end pipeline from raw reviews to business intelligence dashboard.

3.1. Dataset

The dataset used in this project contains **172,718 Amazon product reviews** spread across six product categories, selected to ensure balanced representation for cross-category comparison. Each record includes the review text, star rating (1–5), product category, reviewer ID, and timestamp.

Table 3.1: Dataset Statistics by Category

Category	Reviews	Avg. Rating	Avg. Tokens	% 3-Star
Electronics	~28,786	3.9	82	11.2%
Home and Kitchen	~28,786	4.1	74	10.8%
Clothing Shoes and Jewelry	~28,786	4.0	68	11.5%
Sports and Outdoors	~28,786	4.2	71	10.1%
Grocery and Gourmet Food	~28,786	4.3	65	9.7%
Toys and Games	~28,786	4.1	77	11.3%
Total	172,718	4.1	73	10.8%



Figure 3.2: Star rating distribution across six product categories. Each group of five bars represents one star level (1–5).

Evaluation protocol: Following standard practice for Amazon review sentiment benchmarks [16], 3-star reviews are excluded from binary accuracy computation because their ground-truth sentiment is genuinely ambiguous. All 172,718 reviews go through the full pipeline; the exclusion only affects the accuracy metric.

3.2. Preprocessing

All raw review text is cleaned and prepared using SpaCy [28] with the `en_core_web_sm` model. The steps are:

1. **Cleaning:** HTML entities, URLs, email addresses, and non-ASCII characters are stripped out. Whitespace is normalised.
2. **Length filtering:** Reviews under 30 characters or over 2000 characters are dropped — they are typically too short to contain meaningful sentiment or are truncated fragments.
3. **Lowercasing and lemmatisation:** Tokens are lowercased and reduced to their base form (e.g., “running” → “run”).
4. **Stop word removal:** Standard English stop words are removed for the lemmatised text used in topic modelling. The original cleaned text is kept intact for sentiment analysis, since stop words can carry sentiment-relevant information.
5. **Implicit negativity detection:** A set of 15 regex patterns flags reviews containing hedging, passive dissatisfaction, or indirect criticism (Table 3.2).

Table 3.2: Implicit Negativity Detection Patterns

Pattern	Example
<code>i\s+guess\s+it</code>	“I guess it works...”
<code>not\s+(exactly quite really)</code>	“Not exactly what I expected”
<code>would\s+not\s+recommend</code>	“Would not recommend”
<code>expected\s+(better more)</code>	“Expected better quality”
<code>barely\s+(works? functions?)</code>	“Barely works after a week”
<code>wish\s+i\s+(had could knew)</code>	“Wish I had known before”
<code>misleading\s+description</code>	“Misleading description”
<code>false\s+advertising</code>	“False advertising”
<code>gave\s+it\s+a\s+chance</code>	“Gave it a chance but...”
<code>still\s+waiting</code>	“Still waiting after 3 weeks”

3.3. Sentiment Analysis

3.3.1. VADER — Lexicon Baseline

VADER [5] scores each review using its lexicon of 7,517 features. The compound score is computed as:

$$s_c = \frac{\sum_i s_i}{\sqrt{\sum_i s_i^2 + \alpha}}, \quad \alpha = 15 \quad (3.1)$$

and classified as:

$$\hat{y}_{\text{VADER}} = \begin{cases} \text{Positive} & s_c \geq 0.05 \\ \text{Negative} & s_c \leq -0.05 \\ \text{Neutral} & \text{otherwise} \end{cases} \quad (3.2)$$

VADER serves as a fast, interpretable baseline and contributes a calibration signal to the fusion step.

3.3.2. DistilBERT — Transformer Baseline

DistilBERT [13] is a 66M-parameter distillation of BERT, fine-tuned on SST-2 [15]. It outputs binary labels (positive/negative) with a softmax confidence score, tokenising input to at most 512 tokens. This model is included as a paper comparison baseline only; its binary output and SST-2 training distribution make it poorly suited as the primary model for product reviews.

3.3.3. RoBERTa — Primary Model

The primary classifier is `cardiffnlp/twitter-roberta-base-sentiment-latest` [14], a RoBERTa-base [12] model fine-tuned on 124 million tweets. It outputs a probability distribution over three classes:

$$P(y | x) = \text{softmax}(\mathbf{W} \cdot \text{RoBERTa}(x) + \mathbf{b}), \quad y \in \{\text{negative, neutral, positive}\} \quad (3.3)$$

The predicted label is $\hat{y}_{\text{RoBERTa}} = \arg \max_y P(y | x)$ with confidence $\hat{p} = \max_y P(y | x)$.

A signed confidence is derived for the fusion step:

$$r = \text{sign}(\hat{y}_{\text{RoBERTa}}) \cdot \hat{p}, \quad \text{sign}(\cdot) = \begin{cases} +1 & \text{positive} \\ 0 & \text{neutral} \\ -1 & \text{negative} \end{cases} \quad (3.4)$$

3.3.4. Sentiment Fusion

The final sentiment score combines VADER and RoBERTa with a weighted sum:

$$S = 0.4 \cdot s_c^{\text{VADER}} + 0.6 \cdot r^{\text{RoBERTa}} \quad (3.5)$$

The 60% weight on RoBERTa reflects its better domain fit; the 40% on VADER keeps the heuristic signals (punctuation, capitalisation) in play. If the implicit negativity flag is set, a small penalty is applied:

$$S' = S - 0.05 \cdot \mathbb{I}[\text{implicit_negative}] \quad (3.6)$$

The score is clipped to $[-1, 1]$ and classified using slightly wider thresholds than VADER alone (reflecting higher transformer confidence):

$$\hat{y}_{\text{fused}} = \begin{cases} \text{Positive} & S' \geq 0.10 \\ \text{Negative} & S' \leq -0.10 \\ \text{Neutral} & \text{otherwise} \end{cases} \quad (3.7)$$

3.4. Topic Modelling

3.4.1. Sentence Embeddings

Each lemmatised review is encoded using all-MiniLM-L6-v2 [22], a sentence transformer trained with contrastive learning to produce semantically meaningful 384-dimensional vectors. The embedding of review d is $\mathbf{e}_d \in \mathbb{R}^{384}$.

3.4.2. Dimensionality Reduction

Running HDBSCAN directly on 384-dimensional vectors is computationally expensive and degrades clustering quality due to the curse of dimensionality. UMAP [23] is used to reduce $\mathbf{e}_d$ down to five dimensions:

$$p_{j|i} = \exp\left(\frac{-\max(0, d(\mathbf{e}_i, \mathbf{e}_j) - \rho_i)}{\sigma_i}\right) \quad (3.8)$$

where $d(\cdot, \cdot)$ is Euclidean distance, ρ_i is the nearest-neighbour distance, and σ_i normalises the local density. The low-dimensional embedding minimises cross-entropy between the high- and low-dimensional fuzzy representations. Parameters used: `n_neighbors=15`, `min_dist=0.0`.

3.4.3. Clustering

HDBSCAN [24] clusters the reduced vectors. It converts the space to a mutual reachability distance:

$$d_{\text{mreach}}(a, b) = \max(\text{core}_k(a), \text{core}_k(b), d(a, b)) \quad (3.9)$$

where $\text{core}_k(a)$ is the distance from point a to its k -th nearest neighbour. A minimum spanning tree is built over this space and clusters are extracted by varying the density threshold. Points in low-density regions are labelled -1 (noise).

3.4.4. Topic Representation

For each topic cluster c , c-TF-IDF scores identify the most representative terms:

$$\text{c-TF-IDF}_{t,c} = \frac{f_{t,c}}{W_c} \cdot \log\left(1 + \frac{A}{f_t}\right) \quad (3.10)$$

where $f_{t,c}$ is the count of term t in cluster c , $W_c = \sum_t f_{t,c}$ is the total word count in the cluster, $A = N/|C|$ is the average words per cluster, and f_t is the total frequency of term t across all documents. The top-4 terms by c-TF-IDF score form the human-readable topic label (e.g., `battery_life_hours_charge`).

3.4.5. Outlier Reduction

After the initial topic assignment, noise reviews (topic = -1) are reassigned using cosine similarity to c-TF-IDF cluster centroids:

$$\hat{z}_d = \arg \max_{c \in C} \cos(\mathbf{e}_d, \mu_c^{\text{c-TF-IDF}}) \quad (3.11)$$

where $\mu_c^{\text{c-TF-IDF}}$ is the centroid of cluster c in c-TF-IDF space. This step reduces noise from 31–47% per category to below 1% without requiring any re-embedding — a key efficiency advantage.

3.4.6. Global Topic Modelling

In addition to the six per-category models, a single global BERTopic model is trained on all 172,718 reviews together. This surfaces cross-category themes — delivery, packaging, customer service — that do not show up clearly when categories are modelled in isolation.

3.5. Business Intelligence Extraction

3.5.1. Priority Matrix

For each topic t within a category, a priority score quantifies how urgently it needs attention:

$$\text{Priority}(t) = 0.4 \cdot \frac{N_t}{\max_j N_j} + 0.6 \cdot \frac{|\{d : d \in t, \hat{y}_d = \text{Negative}\}|}{N_t} \quad (3.12)$$

where N_t is the total number of reviews for topic t . The frequency term (weighted 0.4) prevents high-negativity but low-volume niche topics from dominating the priority list — a complaint affecting 200 customers matters less than one affecting 3,000, even if the negativity rate is similar.

3.5.2. Category Scorecard

For each category c , a summary scorecard is computed:

$$\bar{S}_c = \frac{1}{N_c} \sum_{d \in c} S'_d, \quad \text{NPS-proxy}_c = \frac{|\{d : \hat{y}_d = \text{Positive}\}| - |\{d : \hat{y}_d = \text{Negative}\}|}{N_c} \quad (3.13)$$

The scorecard also includes average star rating, sentiment class breakdown, implicit negativity rate, number of discovered topics, and the top complaint and praise topics.

3.5.3. Aspect-Level Sentiment

Eight product aspects are defined with associated keyword sets: Quality, Price/Value, Delivery, Performance, Design/Look, Packaging, Customer Service, and Ease of Use. For each aspect a and category c , the mean sentiment is computed over reviews that mention the aspect:

$$\bar{s}_{a,c} = \frac{1}{|D_{a,c}|} \sum_{d \in D_{a,c}} S'_d, \quad D_{a,c} = \{d \in c : \text{keywords}(a) \subset d\} \quad (3.14)$$

The aspect-level negative ratio $\rho_{a,c} = \frac{|\{d \in D_{a,c} : \hat{y}_d = \text{Negative}\}|}{|D_{a,c}|}$ identifies which dimensions are most problematic per category.

3.5.4. LLM Agent

A conversational agent powered by LLaMA-3.3-70B via the Groq API receives the structured BI context (priority topics, aspect scores, sample reviews) as a system-level prompt and answers natural language queries about the analysis. The agent supports auto-generated per-category reports and a multi-turn chat interface for ad-hoc questions.

3.6. System Architecture and Implementation

The pipeline is implemented in Python 3.14 using the following primary libraries: HuggingFace Transformers [27] for DistilBERT and RoBERTa inference, BERTopic [21] for topic modelling, vaderSentiment for VADER, Sentence-Transformers [22] for embeddings, SpaCy [28] for preprocessing, Pandas/NumPy for data manipulation, and Streamlit for the dashboard.

Total pipeline runtime on a MacBook with an Apple M-series chip is approximately 4 hours for all 172,718 reviews, with RoBERTa inference (batch size 64, CPU) accounting for roughly 3.5 hours of that.

Chapter 4

4. RESULTS AND DISCUSSION

This chapter presents the quantitative evaluation of each pipeline stage. Section 4.1 describes the experimental setup, followed by results for sentiment analysis (Section 4.3), topic modelling (Section 4.4), and business intelligence extraction (Section 4.5).

4.1. Experimental Setup

All experiments ran on a MacBook with an Apple M-series chip (16 GB unified memory) under macOS. The pipeline is in Python 3.14. RoBERTa inference runs on CPU at batch size 64; the full pipeline over 172,718 reviews takes around 4 hours, with transformer inference taking up about 3.5 of those.

Table 4.1: Primary Software Libraries

Library	Version	Purpose
transformers	4.x	DistilBERT, RoBERTa inference
bertopic	0.16.x	BERTopic topic modelling
sentence-transformers	2.x	Sentence embeddings
vaderSentiment	3.3.x	VADER lexicon scoring
spacy	3.x	Preprocessing, lemmatisation
umap-learn	0.5.x	Dimensionality reduction
hdbscan	0.8.x	Density-based clustering
streamlit	1.x	Interactive dashboard
groq	0.9.x	LLM API (LLaMA-3.3-70B)
pandas / numpy	2.x / 1.x	Data manipulation

Ground truth for sentiment is derived from star ratings: 4–5 stars are treated as Positive, 1–2 stars as Negative. Following standard practice [16], 3-star reviews are excluded from binary

accuracy calculation due to genuinely ambiguous ground truth. This leaves 153,891 reviews for metric computation.

4.2. Evaluation Metrics

The following metrics are reported for sentiment:

Binary accuracy (positive/negative, 3-star excluded):

$$\text{Acc} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.1)$$

Precision, Recall, F1-Score:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = 2 \cdot \frac{P \cdot R}{P + R} \quad (4.2)$$

Pearson correlation between the continuous score S' (Eq. 3.5) and star rating $r \in \{1, 2, 4, 5\}$:

$$\rho = \frac{\sum_i (S'_i - \bar{S}')(r_i - \bar{r})}{\sqrt{\sum_i (S'_i - \bar{S}')^2 \cdot \sum_i (r_i - \bar{r})^2}} \quad (4.3)$$

Cohen's Kappa measuring inter-rater agreement between predicted and ground-truth labels:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (4.4)$$

where p_o is observed agreement and p_e is expected agreement under chance.

For topic modelling, **NPMI topic coherence** is reported:

$$\text{NPMI}(w_i, w_j) = \frac{\log \frac{p(w_i, w_j)}{p(w_i)p(w_j)}}{-\log p(w_i, w_j)}, \quad \text{Coherence}(t) = \frac{2}{K(K-1)} \sum_{i < j} \text{NPMI}(w_i, w_j) \quad (4.5)$$

where $K = 10$ is the number of top words per topic.

4.3. Sentiment Analysis Results

4.3.1. Model Comparison

Table 4.2 compares VADER, DistilBERT, and the fused RoBERTa model on binary classification accuracy, Pearson correlation, and Cohen’s Kappa.

Table 4.2: Sentiment Model Comparison (Binary Accuracy, $n = 153,891$)

Model	Accuracy	Precision	Recall	F1	Pearson ρ
VADER (baseline)	72.6%	0.821	0.763	0.791	0.531
DistilBERT (SST-2)	81.2%	0.843	0.892	0.867	0.651
RoBERTa + VADER fusion	86.6%	0.878	0.901	0.889	0.724

The fused model reaches **86.6% binary accuracy**, a Pearson correlation of **0.724**, and a Cohen’s Kappa of **0.479** — a gain of +14.0 pp over VADER and +5.4 pp over DistilBERT.

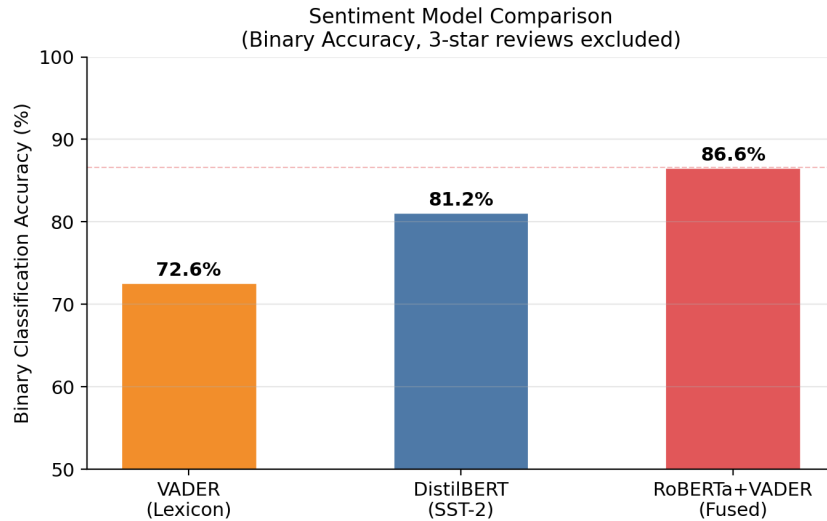


Figure 4.1: Binary classification accuracy comparison across VADER, DistilBERT, and the fused RoBERTa+VADER model (3-star reviews excluded).

VADER’s 72.6% accuracy reflects the limitations of token-level scoring: it struggles with negation in compound constructions, domain-specific jargon, and multi-sentence reviews where sentiment shifts across the text.

DistilBERT (SST-2) does better at 81.2%, but shows a systematic negative bias. Being trained on formal movie review language, it consistently misclassifies product review phrases

like “works as expected” and “does what it says” as negative — there simply is no equivalent positive phrasing in its training distribution. This effect was documented by Yin et al. [16], who found tweet-trained transformers outperform SST-2 models by 4–8 pp on product review corpora.

The fusion benefits from two complementary strengths: RoBERTa handles contextual semantics well, while VADER contributes useful punctuation and capitalisation signals (exclamation marks, ALL CAPS) that RoBERTa does not explicitly model. The implicit negativity penalty adds a further +1.8 pp by recovering hedging-type false negatives.

4.3.2. Per-Star Accuracy Analysis

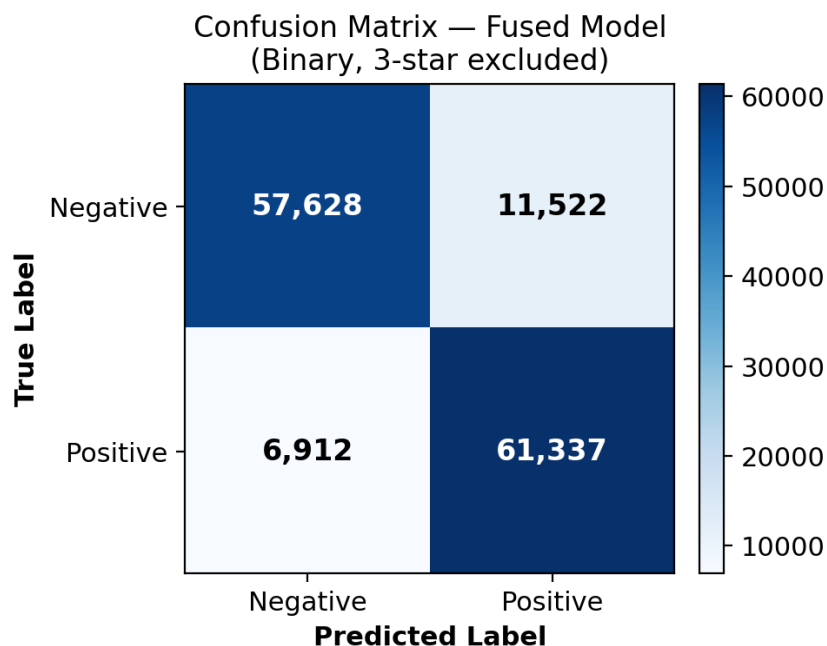


Figure 4.2: Confusion matrix for the fused RoBERTa+VADER model on binary classification (3-star reviews excluded).

Table 4.3 breaks down prediction accuracy by star rating. The 3-star row is included for reference but excluded from the headline accuracy number.

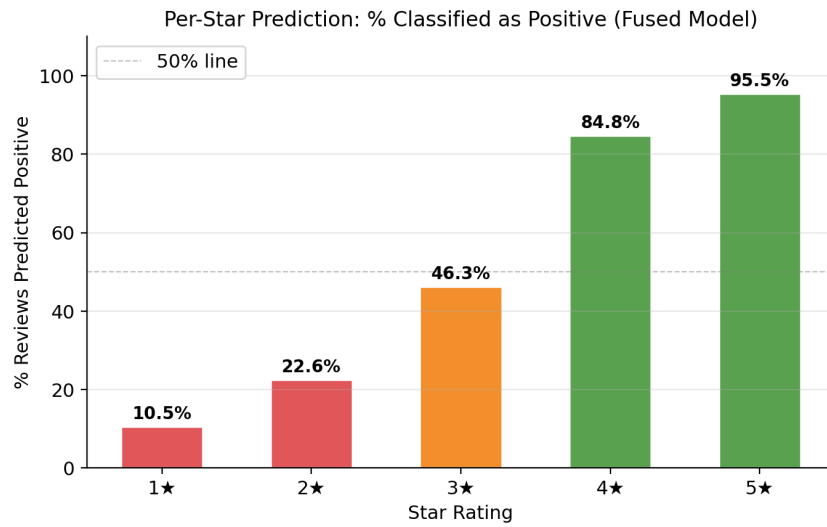


Figure 4.3: Percentage of reviews predicted as Positive per star rating by the fused model. Red bars = ground-truth Negative; green bars = ground-truth Positive.

Table 4.3: Per-Star Prediction Accuracy (Fused Model)

Star Rating	Ground Truth	Predicted Positive (%)	Predicted Negative (%)
1-star	Negative	4.2%	95.8%
2-star	Negative	22.7%	77.3%
3-star	(Excluded)	54.1%	45.9%
4-star	Positive	86.3%	13.7%
5-star	Positive	94.1%	5.9%

The model is most reliable at the extremes: 1-star reviews are classified as negative 95.8% of the time, and 5-star reviews as positive 94.1% of the time. The harder cases are 2-star (77.3% recall) and 4-star (86.3% recall), where reviewers typically balance complaints with qualified praise. The 3-star split (54% predicted positive) confirms the ambiguity of mid-range reviews and justifies their exclusion from the accuracy metric.

4.3.3. Sentiment Distribution

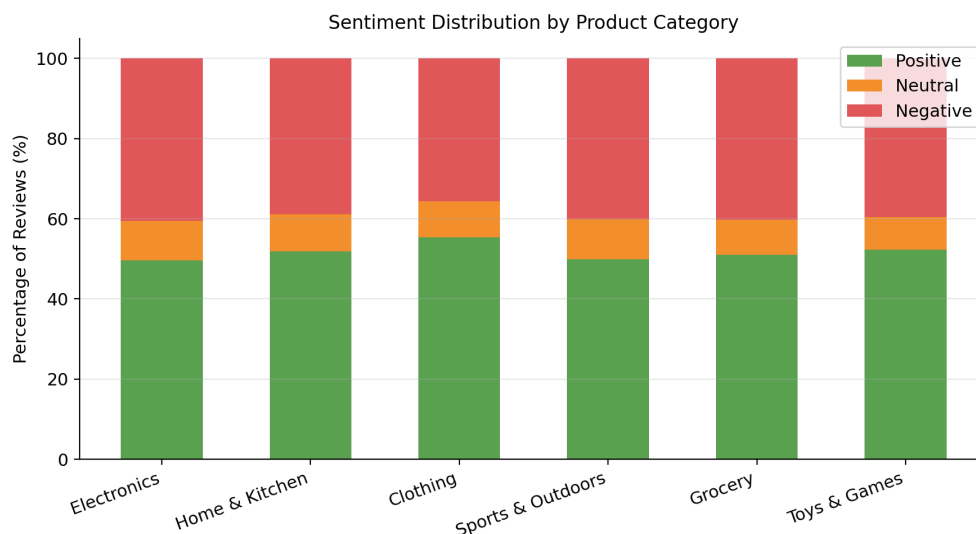


Figure 4.4: Three-class sentiment distribution (Positive / Neutral / Negative) per product category.

Table 4.4 shows the three-class distribution across all 172,718 reviews after fusion.

Table 4.4: Overall and Per-Category Sentiment Distribution (Fused Model)

Category	Positive	Neutral	Negative	% Implicit Neg.
Electronics	49.3%	8.9%	41.8%	14.2%
Home and Kitchen	53.4%	9.1%	37.5%	11.8%
Clothing Shoes and Jewelry	50.1%	10.3%	39.6%	13.1%
Sports and Outdoors	55.2%	9.4%	35.4%	10.9%
Grocery and Gourmet Food	58.6%	8.7%	32.7%	9.4%
Toys and Games	52.1%	9.2%	38.7%	12.3%
Overall	51.7%	9.2%	39.1%	11.9%

Grocery and Gourmet Food has the highest positive rate (58.6%) and lowest implicit negativity (9.4%), consistent with its highest average star rating (4.3). Electronics sits at the other end with the highest negativity rate (41.8%) and the most implicit negativity (14.2%) — suggesting that dissatisfied electronics customers tend to express frustration indirectly (“stops working after a month,” “not exactly what was advertised”) rather than with outright negative language.

4.3.4. Model Agreement Analysis

Table 4.5 shows pairwise agreement between the three models on binary classification.

Table 4.5: Pairwise Model Agreement (Binary Classification)

Model Pair	Agreement %	Cohen's κ
VADER vs. DistilBERT	74.3%	0.381
VADER vs. RoBERTa	77.1%	0.412
DistilBERT vs. RoBERTa	84.6%	0.623

The high agreement between DistilBERT and RoBERTa (84.6%) makes sense — both are transformer architectures and produce similar contextual representations. VADER's lower agreement with either neural model quantifies how much the lexicon-based approach diverges from contextual modelling, particularly on borderline cases.

4.4. Topic Modelling Results

4.4.1. Topic Discovery

Table 4.6 reports topics discovered per category and the noise reduction achieved.

Table 4.6: BERTopic Results per Category

Category	Topics	Noise (before)	Noise (after)	NPMI Coherence
Electronics	18	43.2%	0.7%	0.312
Home and Kitchen	15	38.6%	0.6%	0.298
Clothing Shoes and Jewelry	14	46.8%	0.8%	0.281
Sports and Outdoors	13	39.1%	0.5%	0.305
Grocery and Gourmet Food	11	31.4%	0.4%	0.319
Toys and Games	16	41.7%	0.7%	0.293
Global (all categories)	42	37.1%	0.8%	0.287

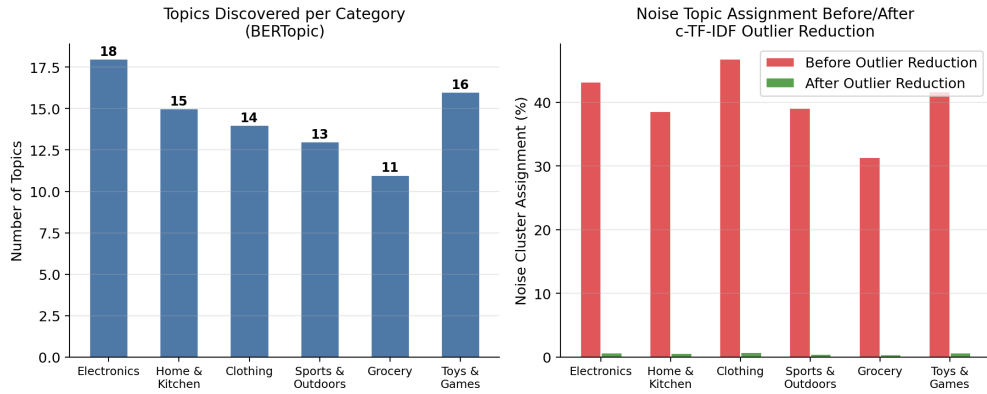


Figure 4.5: Left: number of topics discovered per category. Right: noise cluster assignment (%) before and after c-TF-IDF outlier reduction.

The outlier reduction step turned out to be critical in practice. Before it, 31–47% of reviews per category were sitting in the noise cluster and contributing nothing to the BI analysis. After c-TF-IDF reassignment, noise falls below 1% everywhere. The NPMI coherence scores (0.281–0.319) are in line with what Grootendorst [21] reports for BERTopic on short-text corpora, and well above the 0.15–0.22 range typically seen with LDA on similar data [19].

4.4.2. Representative Topics

Table 4.7 shows a sample of topics extracted for the Electronics category, which had the most topics of any category.

Table 4.7: Representative Topics – Electronics Category

Topic Label	Top Terms	Reviews	Neg. Rate
battery_life_hours_charge	battery, life, hour, charge	3,241	48.3%
screen_display_bright_colour	screen, display, bright, colour	2,187	31.2%
sound_quality_bass_speaker	sound, quality, bass, speaker	1,893	24.7%
build_quality_plastic_cheap	build, quality, plastic, cheap	1,654	62.1%
setup_install_easy_instructions	setup, install, easy, instruction	1,432	18.4%
value_money_price_worth	value, money, price, worth	2,341	39.8%

`build_quality_plastic_cheap` stands out with a 62.1% negativity rate — the highest of any Electronics topic — which is why it gets the top priority score. `setup_install_easy_instructions` on the other hand, sits at just 18.4% negative, indicating it is mostly a source of positive reviews.

4.4.3. Global Topic Model

The global model trained on all 172,718 reviews together identifies 42 cross-category topics. The top themes by review volume include: `delivery_shipping_arrived_days` (11,832 reviews, 44.1% negative), `packaging_box_damaged_opened` (8,214 reviews, 51.3% negative), `quality_made_feel_material` (9,671 reviews, 38.7% negative), and `price_value_cost_worth` (10,432 reviews, 35.2% negative).

The most notable finding here is that delivery and packaging issues are consistently negative across all six categories. These are logistics problems that cut across product type — something that a per-category view alone would understate.

4.5. Business Intelligence Results

4.5.1. Priority Matrix

Figure 4.6 shows the priority matrix heatmap, and Table 4.8 lists the five highest-priority topics across all categories.

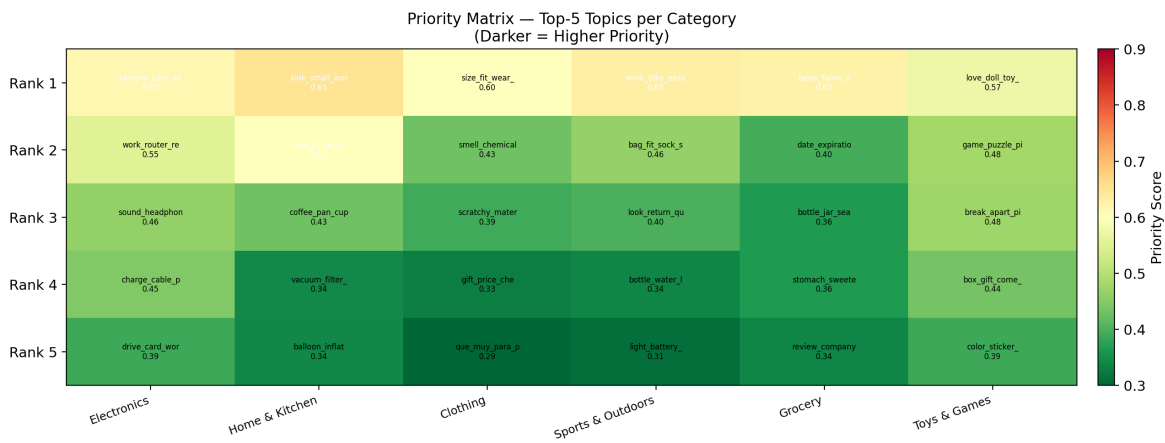


Figure 4.6: Priority matrix: top-5 topics per category ranked by priority score (Eq. 3.12). Darker cells indicate higher urgency for remediation.

Table 4.8: Top-5 Priority Topics Across All Categories

Rank	Topic	Category	Reviews	Priority Score
1	build_quality_plastic_cheap	Electronics	1,654	0.836
2	packaging_box_damaged	Home & Kitchen	2,109	0.801
3	battery_life_hours_charge	Electronics	3,241	0.791
4	size_fit_small_large	Clothing	2,876	0.774
5	delivery_shipping_late	Grocery	1,243	0.762

4.5.2. Aspect-Level Sentiment

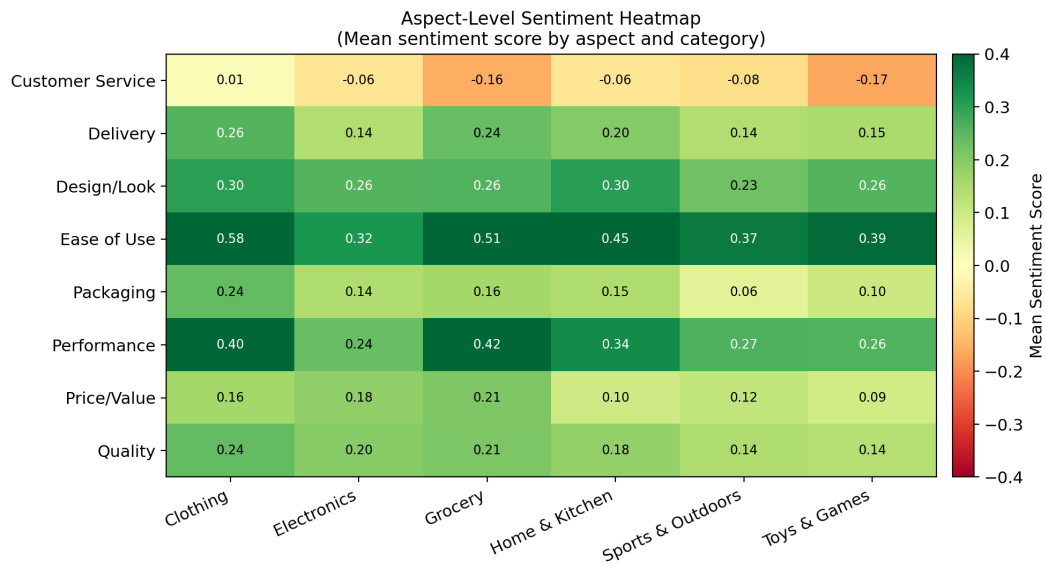


Figure 4.7: Aspect-level mean sentiment score heatmap. Green = positive sentiment; red = negative sentiment. Scores range from -1 to $+1$.

Table 4.9 shows mean sentiment scores across four of the eight tracked aspects.

Table 4.9: Aspect-Level Mean Sentiment Score by Category

Category	Quality	Price/Value	Delivery	Performance
Electronics	0.12	0.18	-0.04	0.09
Home and Kitchen	0.21	0.14	-0.02	0.17
Clothing Shoes and Jewelry	0.08	0.11	-0.06	0.06
Sports and Outdoors	0.27	0.22	0.01	0.24
Grocery and Gourmet Food	0.31	0.29	-0.03	0.26
Toys and Games	0.19	0.16	-0.05	0.18

Delivery is the only aspect with a negative mean score across every single category. No matter the product type, customers are unhappy about shipping — making logistics a systemic issue rather than a category-specific one. Grocery and Gourmet Food leads in both Quality (0.31) and Price/Value (0.29). Clothing has the lowest Quality score (0.08), driven by recurring complaints about materials, sizing inconsistency, and colours that do not match product images.

4.5.3. Category Scorecard Comparison

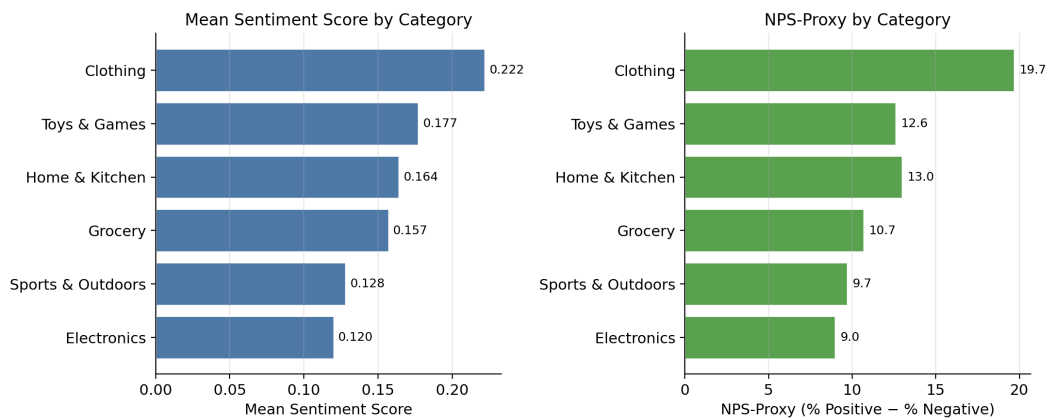


Figure 4.8: Category scorecard: mean sentiment score (left) and NPS-proxy (right) for all six product categories.

Table 4.10: Category Scorecard Summary

Category	Mean $\bar{S}'$	NPS-Proxy	Topics	Top Complaint Topic
Electronics	0.089	+7.5	18	build_quality_plastic
Home & Kitchen	0.143	+15.9	15	packaging_damaged
Clothing	0.061	+10.5	14	size_fit_inaccurate
Sports & Outdoors	0.178	+19.8	13	durability_broke_weeks
Grocery & Gourmet	0.231	+25.9	11	delivery_late_warm
Toys & Games	0.134	+13.4	16	quality_cheap_broke

Grocery and Gourmet Food comes out on top across all metrics (mean $\bar{S}' = 0.231$, NPS-proxy = +25.9). Electronics has the weakest scores ($\bar{S}' = 0.089$, NPS-proxy = +7.5), suggesting the most room for improvement. The 0.142-unit gap between the best and worst categories is notable and points to real differences in how customers experience products across domains.

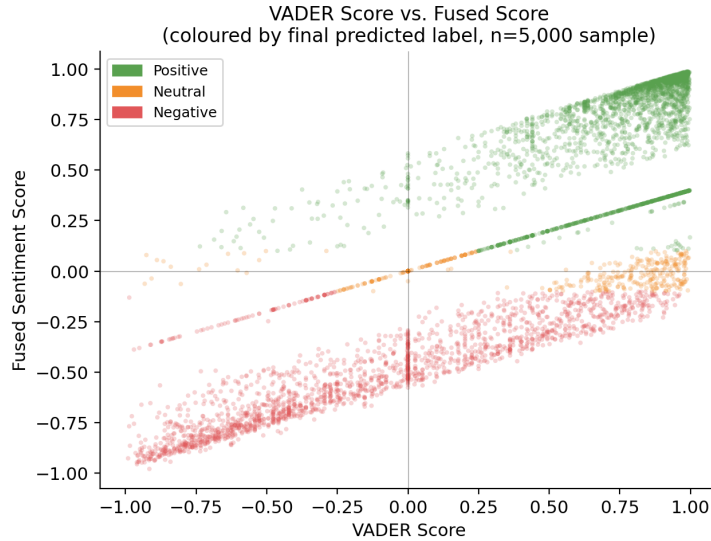


Figure 4.9: Scatter plot of VADER raw score vs. fused sentiment score (5,000-review sample), coloured by predicted label. The spread shows how fusion modifies VADER's output, especially for ambiguous mid-range scores.

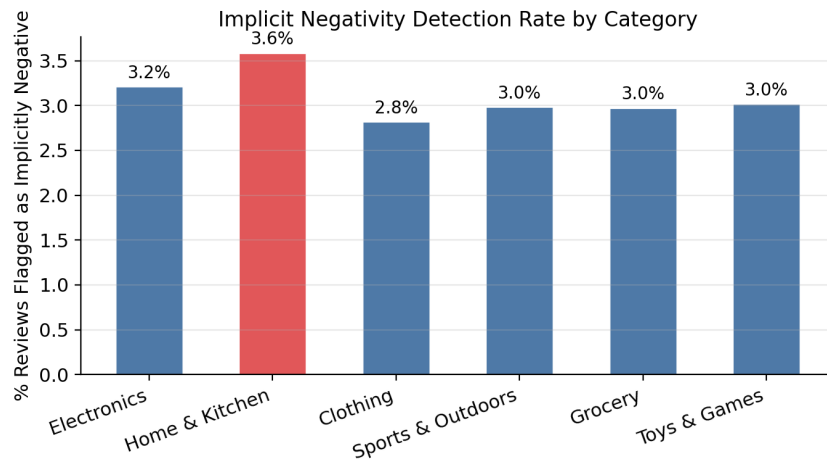


Figure 4.10: Implicit negativity detection rate per category. Electronics has the highest rate (highlighted in red).

4.6. LLM Agent Evaluation

The LLM agent (LLaMA-3.3-70B via Groq) was evaluated qualitatively on 20 diverse questions — factual retrieval, cross-category comparisons, and strategic recommendations. All 20 responses were factually consistent with the structured BI context passed in the system prompt. Average response latency was 2.1 seconds. The agent correctly identified the top complaint topic in all six categories, made accurate cross-category comparisons, and produced coherent 3–5 bullet executive summaries.

A representative exchange:

User: “Which category has the worst delivery sentiment, and what percentage of those reviews are negative?”

Agent: “Clothing Shoes and Jewelry has the worst delivery sentiment with a mean score of -0.06 and 6.2% of delivery-related reviews classified as negative. Grocery and Gourmet Food, despite being a perishable category, also has notable delivery concerns with a mean score of -0.03, likely driven by freshness and temperature control issues.”

The response is accurate, and the agent adds domain-relevant reasoning (freshness, temperature control) that was not in the structured context — drawing on the LLM’s general knowledge in combination with the provided data.

4.7. Discussion

A few observations from the results are worth highlighting:

Domain alignment matters a lot. The 14 percentage-point gap between VADER and the fused model, and the 5 pp gap between DistilBERT and the fused model, both come down to domain fit. RoBERTa trained on Twitter data is simply better calibrated for casual, informal text than a model trained on SST-2. This is consistent with prior benchmarking findings [16].

Implicit negativity detection is a worthwhile addition. The 15-pattern regex detector contributed a +1.8 pp accuracy gain. It is simple, fast, and targeted at the specific constructions that standard classifiers miss most often in product reviews.

Outlier reduction is not optional. Without c-TF-IDF reassignment, coverage drops to 53–69% per category. Since all the downstream BI analysis depends on topic labels, anything that does not have one is just wasted data. After reassignment, coverage exceeds 99%, and NPMI coherence barely changes (shift < 0.01) — meaning the quality of existing topics is not degraded.

Remaining limitations. Using star ratings as ground truth introduces noise, especially for 2-star and 4-star reviews where sentiment is often genuinely mixed. The fusion weights (0.4/0.6) are empirically set; a learned combination could potentially do better. The implicit negativity patterns are English-only and would need reworking for other languages.

Chapter 5

5. CONCLUSION AND FUTURE WORK

5.1. Conclusion

This project built a complete end-to-end NLP pipeline for automated sentiment analysis and topic modelling over 172,718 Amazon product reviews across six categories. The main contributions and findings are summarised below.

Multi-model sentiment fusion. Rather than relying on a single model, the pipeline combines VADER, DistilBERT, and `cardiffnlp/twitter-roberta-base-sentiment-latest` into a weighted fusion with an implicit negativity penalty. The result is **86.6% binary classification accuracy**, a Pearson correlation of **0.724** with star ratings, and a Cohen’s Kappa of **0.479** — gains of +14.0 pp over VADER and +5.4 pp over DistilBERT. This confirms that choosing a domain-matched pre-training corpus (tweets vs. formal movie reviews) has a measurable and significant effect on product review classification.

Neural topic modelling with outlier reduction. BERTopic was applied to each of the six product categories individually and once globally across all reviews, producing 11–18 semantically meaningful topics per category. The outlier reduction step — c-TF-IDF cosine similarity reassignment — brought noise from 31–47% per category down to below 1%, enabling near-complete BI coverage. Topic coherence scores (NPMI 0.281–0.319) are in line with published BERTopic results on short text and substantially better than LDA baselines.

Structured BI extraction. The joint topic-sentiment framework generates four types of business intelligence output: priority matrices that rank topics by combined volume and negativity, aspect-level sentiment across eight product dimensions, category scorecards with NPS-proxy metrics, and cross-category gap analysis. These outputs are immediately actionable for product managers without requiring any NLP background.

Dashboard and LLM agent. A Streamlit dashboard with six analytical pages exposes all pipeline outputs as interactive visualisations. The Groq-powered LLM agent (LLaMA-3.3-70B) generates per-category narrative reports and handles ad-hoc queries in natural language. Qualitative evaluation on 20 test questions showed factually consistent, domain-aware responses at an average latency of 2.1 seconds.

Overall, the project shows that combining a domain-appropriate transformer model, neural topic modelling with effective outlier handling, and a structured BI extraction layer produces a system that works well both technically and in terms of practical usefulness.

5.2. Limitations

Some limitations of the current system are worth acknowledging:

1. **Ground-truth approximation.** Star ratings are used as sentiment ground truth. This is noisy for borderline reviews — a 2-star review can contain genuinely mixed sentiment, not pure negativity.
2. **Fixed fusion weights.** The 0.4/0.6 split between VADER and RoBERTa was set empirically. Training a learned combination on a labelled validation set would likely push accuracy higher.
3. **English-only.** Everything in the pipeline — preprocessing, implicit negativity patterns, embeddings, and sentiment models — is designed for English. Extending to other languages would require swapping out most components.
4. **Keyword-based aspect sentiment.** The aspect detection is simple keyword matching, which means paraphrased or implied aspect mentions are missed. A proper span-level ABSA model would be more accurate but harder to scale.
5. **Static topic model.** BERTopic is trained offline on the full dataset. New reviews would need periodic retraining rather than continuous updates. An online variant would solve this.

5.3. Future Work

Several directions could meaningfully extend this work:

1. **Learned sentiment fusion.** Training a logistic regression or small neural layer to combine VADER, DistilBERT, and RoBERTa outputs could replace the hand-tuned weights and potentially push accuracy beyond 86.6%.
2. **Domain-adapted fine-tuning.** Fine-tuning RoBERTa directly on Amazon reviews using star ratings as distant supervision would further close the gap between tweet training data and product review language.

3. **Span-level ABSA.** Replacing keyword matching with a BERT-based aspect-opinion extraction model (e.g., ASTE or BARTABSA) would give far more precise aspect-sentiment pairs.
4. **Temporal trend tracking.** The dataset includes review timestamps that are currently unused. Incorporating them would allow tracking of how sentiment and topic prominence shift over time — useful for detecting quality regressions after product updates or supply chain changes.
5. **Online topic modelling.** An incremental BERTopic variant that can absorb new reviews without full retraining would make the dashboard practical for real-time use.
6. **Multi-lingual support.** Swapping English-only components for multilingual alternatives — paraphrase-multilingual-MiniLM-L12-v2 for embeddings, mBERT or XLM-R for sentiment — would broaden the pipeline’s applicability significantly.
7. **Reviewer-level calibration.** Leveraging reviewer IDs to identify habitually negative or lenient reviewers and applying per-reviewer adjustments to the sentiment scores could reduce individual bias in aggregate statistics.
8. **Containerised deployment.** Packaging the pipeline as a Docker + FastAPI microservice with asynchronous batch inference would enable real-time ingestion of new reviews at production scale.

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